Mapping the Machine You Live On

COL334 Computer Networks: Assignment 1, Part A

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A report submitted for
COL334: Computer Networks
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August 2026

Abstract

This report covers Part A of COL334 Assignment 1, in which we measure and document the network infrastructure of the IIT Delhi campus from the vantage point of our own devices. Working as cartographers, we traced routes to a fixed set of targets and to a further set of campus and external destinations, isolated the internal (`10.x.x.x`) hops that recur across those traces, and used position, recurrence, and round-trip time to infer the likely role of each hop in the campus topology. We surveyed the wireless layer from three locations, recording every visible SSID, BSSID, band, channel, and signal strength, and used this to determine how many physical access points are actually behind the broadcasts and how the campus's channel plan is enforced. We then ran a set of ping-based delay experiments (fixed-size probes to five targets at increasing distance, a packet-size sweep to isolate transmission delay, and, where data was available, a queue-occupancy comparison between busy and quiet hours) to separate propagation, transmission, and queuing delay in practice. Throughout, we have tried to keep the line between what we directly measured and what we are inferring explicit, and to flag alternative explanations we could not rule out rather than presenting a single tidy story.

Acknowledgements

We thank the COL334 teaching staff for the assignment design and the shared class map, and IIT Delhi's network infrastructure for tolerating two days of low-rate probing.

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Introduction

Assignment 1 for COL334 (Computer Networks) asks every team to map as much of the IIT Delhi campus network as can honestly be measured from an ordinary student device, without authentication attempts, SNMP, vulnerability scanners, aggressive scan timing, or scanning of anyone else's device. This report covers **Part A: Cartography**, worth 50 of the 100 marks for the assignment. Part B (the traffic-capture "stakeout") is being completed and reported separately by our teammate.

0.1 Team

This report was prepared jointly by:

- Tejasvi Shrivastava (2024CS10582)
- Yashvardhan Singh Chaudhary (2024CS10300)

0.2 Rules of Engagement

All measurements in this report were taken under the constraints set out in the assignment handout: no authentication attempts, no SNMP, no exploits or vulnerability scanners, scan timing no more aggressive than `-T2`, and no scanning of devices that might belong to a student or staff member rather than to the infrastructure itself.

0.3 Fixed Targets

Wherever this report refers to "the targets", it means the five fixed destinations specified in the handout:

- **T1**: our default gateway (found per-device, see Chapter 1)
- **T2**: `www.iitd.ac.in`
- **T3**: `8.8.8.8`
- **T4**: `www.iitb.ac.in` (~1100 km)
- **T5**: `www.mit.edu` (~12,000 km)

0.4 Chapter List

Chapter 1, Your Path Out. Our own device-to-Internet path: interface, gateway, DNS, and the first hops out of campus, each tagged with evidence.

Chapter 2, Campus Topology. The router hunt (A2.1), the wireless survey (A2.2), and our contribution to the shared class map (A2.3).

Chapter 3, The Delay Experiment. Where RTT time goes (A3.1), making transmission delay visible via a packet-size sweep (A3.2), and watching a queue fill (A3.3).

Chapter 4, Evidence. The numbered evidence items (E-1 onward) that the rest of the report cites.

Chapter 5, Findings. Three specific, checkable findings about the network, each with an alternative explanation we could not rule out.

0.5 A Note on Honesty

The assignment explicitly rewards the distinction between what we observed directly and what we are inferring, and rewards an honestly reported "no difference" or "couldn't reach" over a tidied-up positive result. We have tried to keep to that standard throughout: flagging inferred elements, reporting dropped probes as data rather than omitting them, and naming a second explanation wherever our evidence does not fully rule one out.

Chapter 1

Your Path Out

This chapter documents the one part of the network we can see completely: the path from our own device out to the open Internet.

1.1 Device and Access Point

Measurements were taken from both teammates' devices, both in Satpura Hostel but in different rooms. Tejasvi's laptop (Ubuntu) connects over Wi-Fi (interface `wlo1`) from room NB-09; the nearest access point is photographed in Evidence E-1 (Figure 1.1). Yashvardhan connects from room NB-01, also Satpura Hostel; his nearest access point is photographed in Evidence E-2 (Figure 1.2). Both photos carry a handwritten placard giving roll numbers, date, and location.

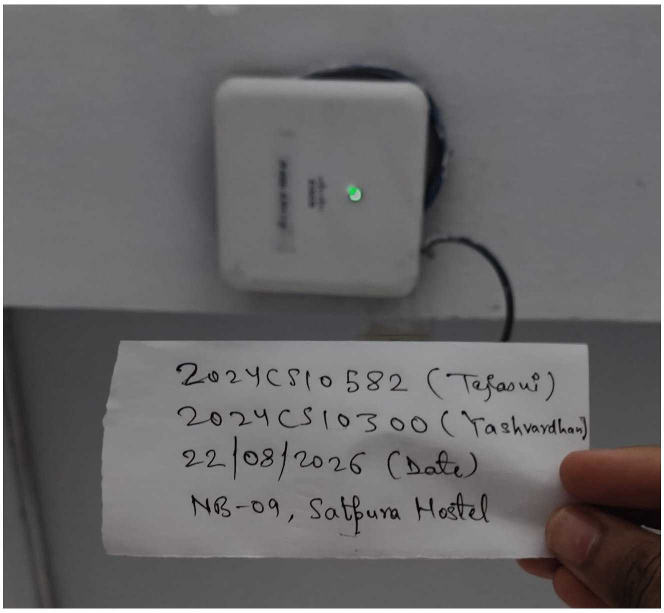


Figure 1.1: The access point/wall unit nearest Tejasvi's device (NB-09) during measurement, with evidence placard. See Evidence E-1.

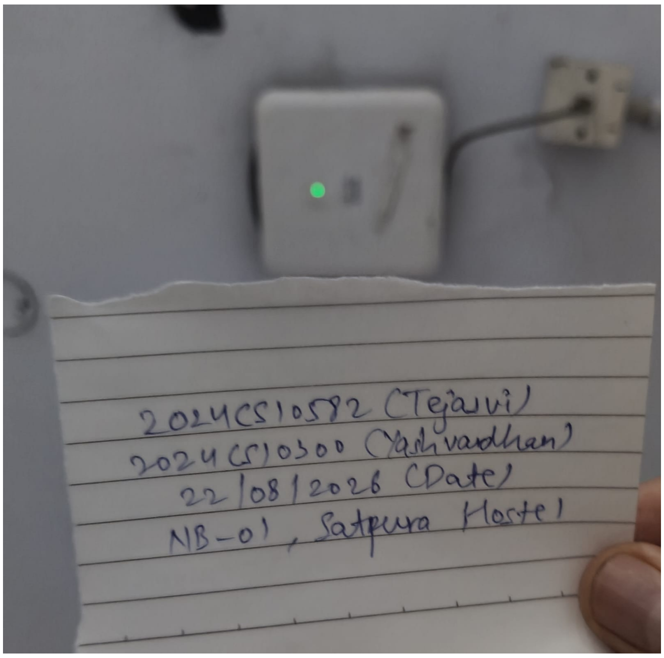


Figure 1.2: The access point/wall unit nearest Yashvardhan's device (NB-01, same hostel), with evidence placard. See Evidence E-2.

1.2 Finding T1

`ip -br a` shows our device's address as `10.184.6.40/19` on `wlo1`. `ip route | grep default` gives:

```
default via 10.184.0.1 dev wlo1 proto dhcp src 10.184.6.40 metric 600
```

so the DHCP-assigned default gateway is **10.184.0.1**. `resolvectl status` shows our active DNS resolver as **10.7.172.202** (with 10.10.1.4, 10.10.1.5, 10.10.2.2 also configured).

A discrepancy worth flagging. The handout notes that a traceroute to anything outside campus should show the gateway at hop 1 "by definition", and that if it doesn't, that's worth a sentence. In our case it doesn't: every traceroute we ran (Chapter 2, and the one below) shows hop 1 as **10.184.0.13**, not 10.184.0.1. Both addresses are in the same `/19`, so our best explanation is that 10.184.0.1 is a virtual/first-hop address (e.g. a shared gateway IP fronting several access switches, possibly via VRRP-style redundancy), while 10.184.0.13 is the specific physical router interface that actually decrements TTL and replies with the ICMP time-exceeded message, i.e. our packets' first real Layer-3 hop is a different, specific box behind the advertised gateway address. We cannot fully rule out the alternative that 10.184.0.1 simply never responds to traceroute probes (rate-limited or filtered ICMP) while still being the true first hop; we have no way to distinguish "different device" from "same device, doesn't reply on this address" without further tooling that is out of scope here. For the rest of this report we follow the handout's implicit intent and use **10.184.0.13** as T1, since that is the address that is consistently and observably our first hop (Evidence E-5; Yashvardhan's equivalent `ip route`/traceroute output, Evidence E-6, confirms the same address).

1.3 Path to the Boundary

A traceroute to an external destination not previously used elsewhere in this report (`www.codechef.com`, chosen to avoid double-counting a target already used for T1–T5) gives (Evidence E-7):

```
 1  10.184.0.13      4.929 ms  4.856 ms  4.816 ms
 2  10.255.109.100   4.761 ms  4.732 ms  4.702 ms
 3  10.255.107.3     4.679 ms  4.649 ms  4.622 ms
 4  10.119.233.65    4.590 ms  4.560 ms  4.531 ms
 5  * * *
 6  10.119.234.162   8.483 ms  7.723 ms  7.581 ms
 7  * 14.143.159.145 11.753 ms *
 8  * * *
 9  121.240.255.30   11.581 ms  11.539 ms *
10  104.23.231.30 17.261 ms * 104.23.231.11 9.883 ms
```

```
11  172.67.75.52     9.713 ms  8.882 ms  8.770 ms
```

Hop 5 and hop 8 are unresponsive; hop 7 is the first public IP (first boundary crossing); hop 11 is the destination itself. Note that hop 10 returns two different IP addresses on the same line (`104.23.231.30` and `104.23.231.11`), indicating the upstream network is actively load-balancing our ICMP probes across multiple Cloudflare edge servers via Equal-Cost Multi-Path (ECMP) routing, rather than all three probes hitting the same device.

Hops 1–6 are all internal (`10.x.x.x`) and match the core/egress path already identified in Table 2.1 (Chapter 2): local gateway, the two core distribution routers, the internal border router, and the external egress gateway. Hop 5 does not respond, which we read as an unresponsive internal hop rather than a gap in the path, since hop 6 replies with a plausible, monotonically-consistent RTT. The first *public* IP appears at hop 7:

Table 1.1: Whois lookups for the first public hops after our gateway.

Hop	IP	Owner (whois)	Role
7	14.143.159.145	Tata Communications (TATACOMM-IN)	Upstream ISP hop
9	121.240.255.30	Tata Communications (TATACOMM-IN, ex-VSNL)	Upstream ISP hop
10	104.23.231.30 / .11	Cloudflare, Inc. (CLOUDFLARENET)	CDN edge

This is where our traffic crosses the IIT Delhi boundary: hop 6 (10.119.234.162, the internal egress gateway) hands off to hop 7, the first Tata Communications address. From there it stays on Tata's network (hop 9) before landing on a Cloudflare edge node (hop 10) that terminates `www.codechef.com`, consistent with the CDN-edge explanation we gave in Chapter 3 for why T5's measured minimum RTT undercut its nominal geographic distance: Tata Communications appears to be (at least one of) our commercial upstream(s), and a lot of what looks like "the open Internet" from here is actually a nearby CDN edge rather than the origin server.

1.4 Diagram: Device to Internet

Figure 1.3 draws both teammates' paths from device to the open Internet, merging at the shared gateway, following the internal core/egress path, crossing the IIT Delhi boundary, and stopping at a dashed line past which everything is inferred rather than directly observed.

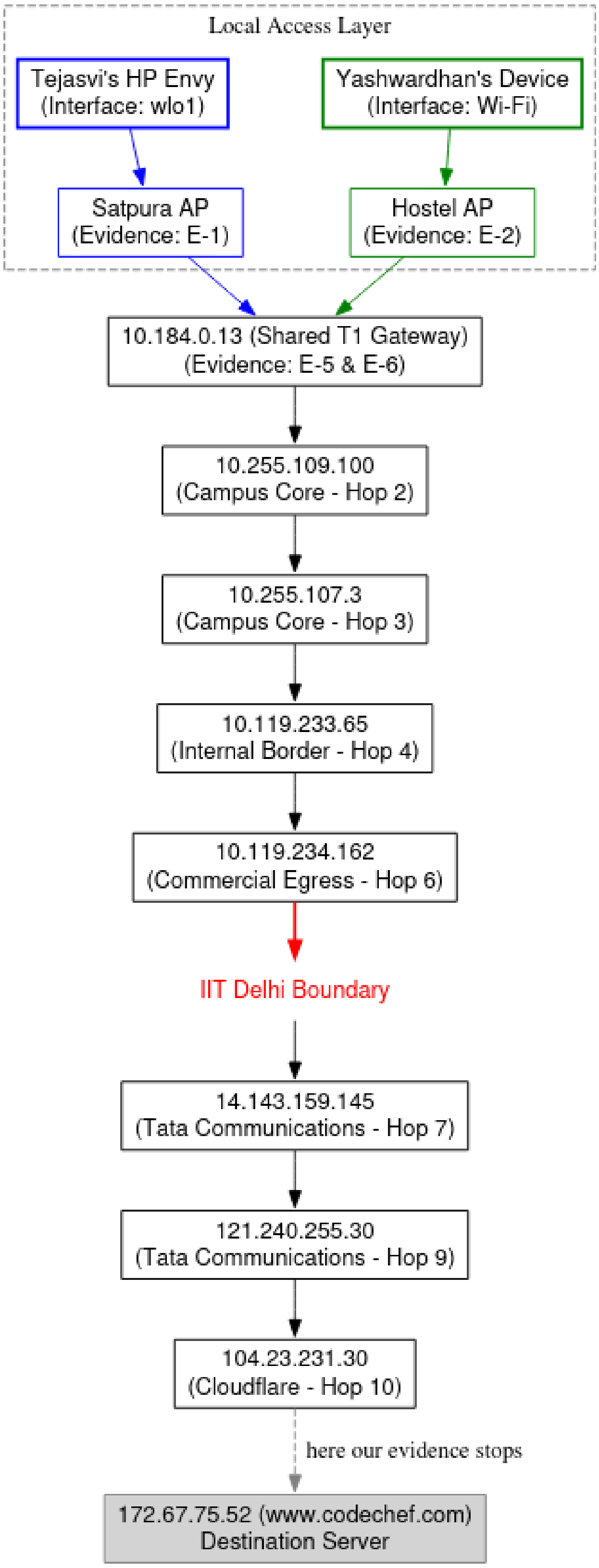


Figure 1.3: Both teammates' device-to-Internet paths, merging at the shared T1 gateway. Evidence tags reference Chapter 4; everything past the dashed line is inferred, not directly observed.

Analytical notes. Two things in this diagram are worth spelling out rather than leaving implicit:

The shared gateway. Both teammates' independent traceroutes hit **10.184.0.13** at hop 1 (Evidence E-5, E-6), even though they capture from different access points in different rooms (NB-09 and NB-01). Since both rooms are in the same hostel (Satpura), this is a fairly mild result on its own: one first-hop gateway serving one hostel building is the ordinary case, not a surprising one. What it does confirm concretely is that IIT Delhi is not assigning a separate default gateway per room or per AP within a hostel; the access-layer merge happens at (at least) the building level. We cannot rule out the alternative that 10.184.0.13 in fact serves a wider area than just Satpura (multiple hostels sharing one gateway); distinguishing the two would need a third vantage point in a different hostel that still resolves to the same first hop.

The "missing" hops. Hops 5 and 8 in Section 1.3's traceroute returned `* * *` (request timeouts). We read this as *not* a routing failure: the surrounding hops (4 and 6; 7 and 9) all reply with plausible, monotonically-increasing RTTs, which would not happen if a hop in between were actually dropping our packets. The more likely explanation is that these specific routers are configured not to generate ICMP time-exceeded replies at all (a common practice for masking internal infrastructure), meaning our packets successfully transited them on the way to the next hop, but the routers themselves declined to identify themselves. The alternative we can't fully rule out is transient packet loss specific to the one probe we sent per hop; re-running the traceroute with more probes per hop would help distinguish "never replies" from "dropped that particular probe".

1.5 Evidence Tags

Every element in the final diagram must carry an evidence tag (E-1, E-2, . . .) referencing Chapter 4; untagged elements score zero per the assignment's marking rules.

Chapter 2

Campus Topology

2.1 A2.1: The Router Hunt

Traceroutes were run to at least 10–15 destinations: campus services (`www.iitd.ac.in`, `moodle`, `library`, `webmail`, `cse`, `am`, `chemical`, `design`, `dms`, `textile`, `ee`, `maths`) and external destinations (`1.1.1.1`, `8.8.8.8`, `www.iitb.ac.in`, `www.mit.edu`, and several large commercial sites used purely to sample more of the egress path). Table 2.1 lists every distinct internal (`10.x.x.x`) hop observed, an abbreviated view of which traces it appeared in, its position along the path, its average RTT, and our best guess at its role, reasoned from that position, recurrence, and RTT together.

Table 2.1: Observed internal hops and inferred roles. "Traces" is abbreviated to a representative subset with a count; the full per-trace list for each hop is preserved as Evidence E-2.

Hop IP	Representative traces (count)	Pos.	Avg RTT (ms)	Best guess at role
10.184.0.13	www.iitd.ac.in, moodle, cse, 8.8.8.8, www.iitb.ac.in, … (24)	1	4.45	Local gateway (T1) for the hostel area
10.254.236.26	am, webmail, textile, dms, moodle, www.iitd.ac.in, design (7)	2	5.09	Primary aggregation router (campus services)
10.254.236.6	ee, library, chemical, maths (4)	2	4.55	Secondary aggregation router
10.10.17.1	moodle (1)	3	3.62	Department / service subnet gateway
10.254.208.6	cse (1)	2	7.17	Department subnet gateway (CSE)
10.208.20.2	cse (1)	3	7.17	Department subnet gateway (CSE, internal)
10.10.211.213	am (1)	3	8.44	Department / service subnet gateway
10.7.10.101	chemical (1)	3	6.22	Department / service subnet gateway
10.10.17.2	design (1)	4	6.13	Department / service subnet gateway
10.10.211.211	textile (1)	3	7.51	Department / service subnet gateway
10.10.211.216	maths (1)	3	4.96	Department / service subnet gateway
10.255.109.100	whatsapp, amazon, instagram, google, netflix, mit.edu, … (15)	2	5.01	Campus core distribution router
10.255.107.3	(as above) + www.iitb.ac.in (16)	3	5.39	Campus core distribution router
10.119.233.65	(as above) (16)	4	5.99	Internal border / transit router
10.119.234.162	amazon, ibm, netflix, mit.edu, wikipedia, x, microsoft, reddit, apple (9)	6	8.55	External egress gateway (commercial ISP path)
10.1.207.69	whatsapp, instagram, iitb.ac.in, google, facebook, 8.8.8.8 (6)	5	29.22	Transit router (NKN / peering path)
10.1.200.137	whatsapp, instagram, iitb.ac.in, google, facebook, 8.8.8.8 (6)	6	34.58	Transit router (NKN / peering path)

Note: position/RTT reconstructed from a flattened source table; verify against the original CSV before final submission (likely viva material).

A few observations follow directly from the table. `10.184.0.13` appears at position 1 in every trace we ran, consistent with it being our own default gateway rather than a shared campus device. The pair `10.255.109.100` / `10.255.107.3` / `10.119.233.65` appears in every trace to an external commercial destination but never in traces to `www.iitd.ac.in` or other campus services, which is the signature of a core/egress path that campus-internal traffic never needs to cross. Two hops, `10.1.207.69` and `10.1.200.137`, sit at a noticeably higher average RTT (29–35 ms) than the rest of the internal hops (4–9 ms); combined with their appearing only on traces toward external destinations and not toward `www.iitb.ac.in` exclusively, we read them as the peering/transit hops just before or on IIT Delhi's connection to the National Knowledge Network (NKN) or a commercial upstream, rather than anything internal to a single building. The department gateways (`10.10.17.x`, `10.10.211.x`, `10.208.20.2`, `10.7.10.101`, `10.254.208.6`) each appeared in exactly one trace, which is expected: each department's traceroute only crosses its own subnet gateway.

2.1.1 Internal Topology Graph

Figure 2.1 draws the internal topology as a graph, with routers as nodes and observed adjacencies (which hop directly followed which, across all traces) as edges.

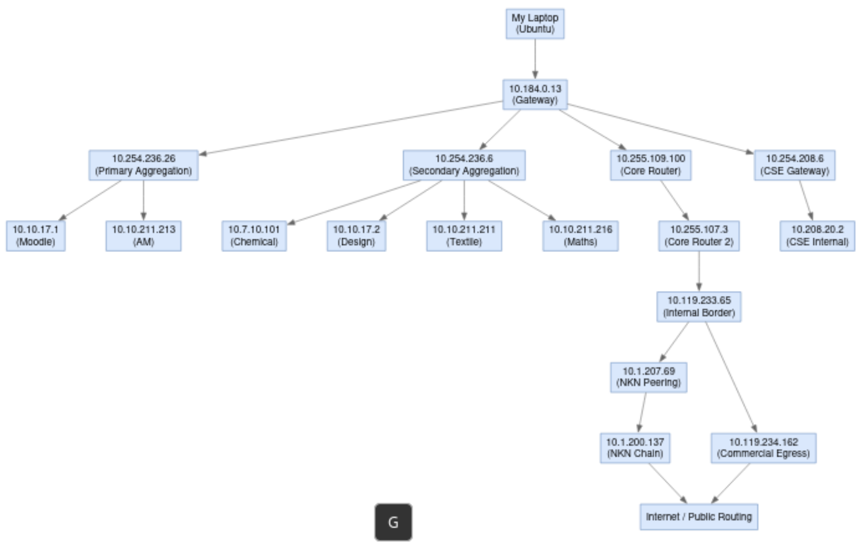


Figure 2.1: Internal topology graph built from observed adjacencies in Table 2.1. Edges represent a hop directly following another in at least one trace; nothing past the last labelled node was directly observed.

2.2 A2.2: The Wireless Layer

A Wi-Fi survey was run from three locations on campus using `nmcli`, recording every broadcast SSID together with its BSSID, band, channel, and signal strength.

Table 2.2: Wireless survey across three locations (redundant enterprise BSSIDs at Satpura truncated for length; full capture preserved as Evidence E-3).

Location	BSSID	SSID	Band	Ch.	Sig.
Satpura	4E:BC:FE:A5:A4:A4	realme P2 Pro 5G	2.4 GHz	6	97
Satpura	E4:4E:2D:0C:82:61	eduroam	2.4 GHz	1	74
Satpura	E4:4E:2D:0C:82:60	IITD_WIFI	2.4 GHz	1	74
Satpura	E4:4E:2D:0C:82:64	IITD_Secure_GUEST	2.4 GHz	1	74
Satpura	E4:4E:2D:0C:82:63	(Hidden)	2.4 GHz	1	74
Satpura	E4:4E:2D:0C:82:6E	eduroam	5 GHz	44	70
Satpura	E4:4E:2D:0C:82:6F	IITD_WIFI	5 GHz	44	69
Satpura	E4:4E:2D:0C:8B:E1	eduroam	2.4 GHz	6	59
Satpura	A4:B4:39:40:C3:61	eduroam	2.4 GHz	11	59
Satpura	A4:B4:39:33:7E:A4	IITD_Secure_GUEST	2.4 GHz	11	40
Satpura	88:9C:AD:5A:43:41	eduroam	2.4 GHz	11	29
Lecture Hall	54:8A:BA:DD:7A:C1	IITD_Secure_GUEST	2.4 GHz	6	97
Lecture Hall	54:8A:BA:DD:7A:C4	IITD_WIFI	2.4 GHz	6	87
Lecture Hall	54:8A:BA:DD:7A:C3	IITD_IOT	2.4 GHz	6	87
Lecture Hall	54:8A:BA:DD:7A:CE	IITD_Secure_GUEST	5 GHz	36	82
Lecture Hall	04:5F:B9:2F:24:C1	IITD_Secure_GUEST	2.4 GHz	1	69
Lecture Hall	70:BD:96:6E:75:42	eduroam	2.4 GHz	11	65
Lecture Hall	38:7A:CC:6D:56:A9	3E4C54F42B90	5 GHz	44	59
Lecture Hall	78:22:88:40:9F:42	454B10749644	5 GHz	44	44
Lecture Hall	04:5F:B9:2F:24:CB	IITD_WIFI	5 GHz	116	42
Library	70:DF:2F:04:98:62	eduroam	2.4 GHz	1	72
Library	70:DF:2F:04:98:63	IITD_IOT	2.4 GHz	1	72
Library	5C:A6:2D:44:E7:E2	eduroam	2.4 GHz	11	62
Library	5C:A6:2D:44:E7:ED	eduroam	5 GHz	128	56
Library	70:DF:2F:04:98:6C	IITD_IOT	5 GHz	149	52
Library	B8:B7:DB:E6:BD:B5	26582626	2.4 GHz	5	42
Library	BE:00:19:A0:0C:FA	rjnish	2.4 GHz	6	40
Library	5C:A6:2D:6F:F8:E1	IITD_Secure_GUEST	2.4 GHz	1	39

2.2.1 Reading the Survey

How many physical APs. SSIDs are not access points: a single radio commonly broadcasts several SSIDs from one MAC block (multiple BSSID / MBSSID), differing only in the last hex digit. Grouping BSSIDs by their shared prefix rather than counting SSID rows gives 5 distinct physical enterprise APs in Satpura, 3 in the Lecture Hall, and 4 in the Library, noticeably fewer than the raw broadcast count.

Global vs. localised SSIDs. `eduroam`, `IITD_WIFI`, and `IITD_Secure_GUEST` appear in all three locations and are evidently provisioned campus-wide. `IITD_IOT` appears in both academic locations (Lecture Hall, Library) but not once in the Satpura data, which reads as a deliberate choice to keep IoT infrastructure out of residential buildings rather than a coverage gap, though we cannot rule out that our single Satpura vantage point simply missed an IITD_IOT radio elsewhere in the hostel. Personal hotspots (`realme P2 Pro 5G`, `rjnish`, the two unnamed hex-string SSIDs in the Lecture Hall) are, as expected, seen at exactly one location each and nowhere else.

Channel plan. Across all three surveys, zero networks were observed on 2.4 GHz channels 3, 4, or 8: every enterprise AP sits on 1, 6, or 11, the three mutually non-overlapping channels. This is consistent with the campus enforcing a fixed channel plan (RRM) on managed infrastructure. The one exception is the student hotspot `26582626` in the Library, broadcasting on channel 5, which does create adjacent-channel interference with neighbouring channel-1 and channel-6 networks, but since it is a personal device we cannot scan or otherwise investigate further under the assignment's rules.

2.3 A2.3: The Class Map

Our router-hunt and wireless-survey data (Sections 2.1–2.2) were added to the shared `COL334_A1_Campus_Map.xlsx`, and we independently verified two rows submitted by another team.

Contributions. We contributed 33 previously-undocumented infrastructure IPs to the shared map (Evidence E-3, E-4, which already cover the underlying wireless-survey and router-hunt data this was built from): the internal aggregation-layer routers already listed in Table 2.1 (e.g. `10.254.236.x`), plus a further set of external transit and CDN-edge addresses seen only on paths to non-campus destinations, including NKN peering (`10.1.200.137`), Meta's peering/transit edge, and various CDN edges for `mit.edu` and `google.com`, among others. These were left out of Table 2.1 itself to keep it focused on the destinations this report discusses in depth, but they are now part of the class-wide picture.

Verification. We selected two rows submitted by another team (2024CS50261+2024CS51073) to check independently from our own vantage point:

- Row 1092: `10.255.221.1`, claimed at hop 11.
- Row 1096: `10.255.237.94`, claimed at hop 7.

Both came back **couldn't reach**: a fresh `traceroute -n` and `ping -c 10` to each IP, run directly from the Satpura access layer, returned 100% timeouts/packet loss across all 30 traceroute hops and all 10 pings (Evidence E-15). We read this as more likely an architectural access-control choice than a sign the other team's row is wrong: these could be core routers that simply do not answer unsolicited ICMP originating from student-subnet IP space at all. That would be distinct from the hop 5/hop 8 case in Chapter 1, which only suppressed TTL-exceeded replies while still forwarding transit traffic; here nothing came back at all, including to packets that were never expected to traverse these IPs as intermediate hops. The other explanation we can't rule out is more mundane: the other team's claimed hop-position or IP could simply be a typo or a stale reading from when that address briefly appeared in one of their traces, in which case "couldn't reach" would reflect an error in the source row rather than anything about the router's configuration. We have no way to distinguish the two without asking the submitting team what they saw at the time.

Chapter 3

The Delay Experiment

3.1 A3.1: Where Does the Time Go?

Each of T1–T5 was pinged 50 times; Table 3.1 records the minimum, average, and maximum RTT observed for each. Distances are rough straight-line estimates (T2/T4 from map distance to the named campus, T3/T5 from public IP geolocation, i.e. the resolved IP's registered location) and are not independently verified against ground truth.

Table 3.1: Aggregate RTT statistics, 50 packets per target.

Target (role)	Resolved IP	Est. distance	Min RTT	Avg RTT	Max RTT
T1 (local gateway)	10.184.0.13	<0.1 km	3.904	8.891	96.290
T2 (campus target)	www.iitd.ac.in	∼2.0 km	3.315	7.489	14.110
T3 (regional DNS)	8.8.8.8	∼20.0 km	42.824	49.859	119.830
T4 (IIT Bombay)	www.iitb.ac.in	∼1100 km	N/A: 100% packet loss		
T5 (global target)	www.mit.edu	nominal 12,000 km	35.596	51.145	247.982

All RTT values in ms.

T4 returned 100% packet loss across all 50 probes. We read this as ICMP being firewalled at or near `www.iitb.ac.in` rather than a routing failure, since traceroutes toward the same destination (Chapter 2) do get responses from intermediate hops, which would not happen if the path itself were broken.

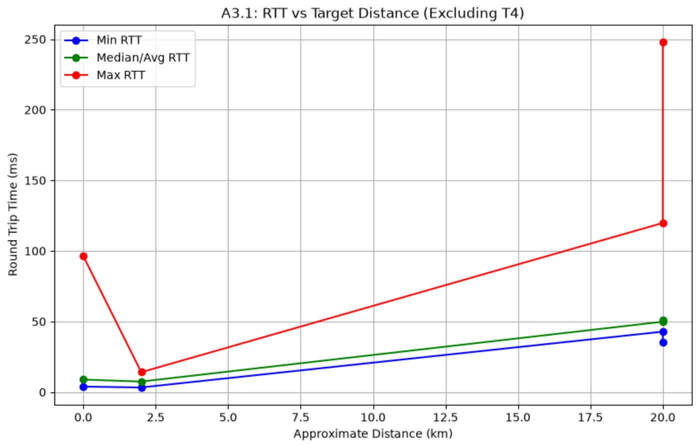


Figure 3.1: Minimum, median/average, and maximum RTT against approximate distance for T1, T2, T3, T5 (T4 excluded, no response).

Propagation delay vs. measured minimum. At 2×10^8 m/s, T2's ~2 km distance implies a propagation delay of about 0.02 ms, negligible next to the measured 3.3 ms minimum, so for a target this close essentially all of the RTT floor is transmission and processing delay, not propagation. T5's 12,000 km distance implies a propagation floor of about 120 ms round-trip, yet the measured minimum was 35.6 ms, well under that floor. The straightforward reading is that our traffic did not actually travel the full nominal distance: `www.mit.edu` is very likely served to us from a CDN edge node much closer than Massachusetts. An alternative we cannot fully rule out is that the geolocation-based distance estimate itself is simply wrong (IP geolocation for large anycast/CDN-fronted services is frequently inaccurate), in which case the "true" distance to whatever server actually answered us is just smaller than 12,000 km rather than the packet having taken an implausible shortcut. Either way, the number confirms we are not talking to a server in Cambridge, MA.

Min-to-max/average gap is queuing delay. Propagation and transmission delay for a given path and packet size are effectively fixed, so the spread between the minimum and the average/maximum RTT for the same target is attributable to queuing delay: time spent sitting in router buffers behind other traffic. This is why it shows up as *variation* rather than a constant offset: queue occupancy depends on how much cross-traffic happens to be competing for the same link at the moment our packet arrives, which changes probe to probe. T3 shows the widest such gap (42.8 ms min vs. 119.8 ms max), consistent with a longer, shared path through more congested infrastructure than the one-hop path to T1.

T1's non-zero RTT. T1 is one hop away, yet its minimum RTT is 3.9 ms, not 0.

Over Wi-Fi, this floor is built from MAC-layer overhead we do not see in the ICMP timestamp: CSMA/CA clear-channel assessment before the medium can be used, the mandatory DIFS/backoff interval, and (in both directions) the transmission delay of putting the frame on and off the physical layer, plus whatever small forwarding/processing delay the gateway itself adds. None of this is propagation delay: at <0.1 km, propagation is on the order of nanoseconds.

3.2 A3.2: Making Transmission Delay Visible

T1 and T3 were pinged at packet sizes 64, 256, 512, 1024, and 1400 bytes (30 packets each); the minimum RTT at each size was plotted against packet size and fit with a straight line for each target (Figure 3.2).

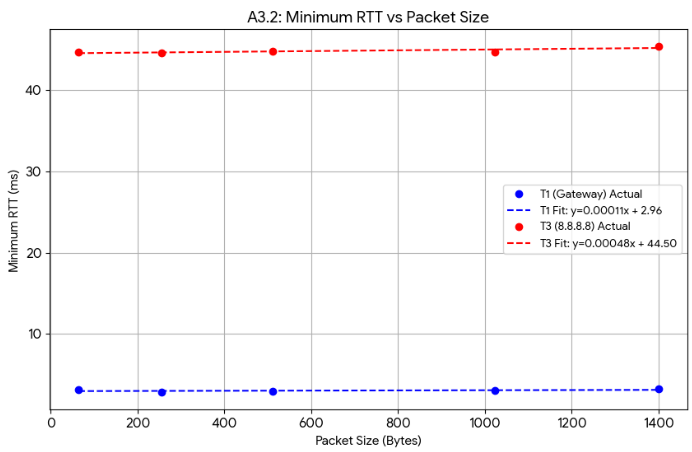


Figure 3.2: Minimum RTT against packet size for T1 and T3, with linear fits.

Fitted lines: T1, $y = 0.00011x + 2.96$; T3, $y = 0.00048x + 44.50$ (RTT in ms, x in bytes).

The intercept. The y-intercept is the extrapolated RTT for a 0-byte payload, i.e. everything that is *not* transmission delay for the payload itself: propagation delay, router/host processing, and (for T1) Wi-Fi contention overhead. T3's much larger intercept (44.5 ms vs. 2.96 ms) reflects the extra propagation and per-hop processing accumulated over the longer, multi-hop path to 8.8.8.8, versus the single Wi-Fi hop to T1.

The slope. The slope is transmission delay per byte, L/R, and since a multi-hop path uses store-and-forward switching, the effective slope is the *sum* of $1/R_i$ over every link on the path, not just the first one. T1's slope (0.00011 ms/byte) is close to flat because one Wi-Fi hop's link rate is high enough that transmission delay for a payload in this size range barely moves the RTT. T3's slope (0.00048 ms/byte) is roughly four times steeper, which is consistent with the packet crossing several additional links (campus core, egress, upstream) each contributing their own L/R_i term, rather than transmission delay being determined by a single bottleneck link. We note that both slopes are small enough, and the underlying ping resolution coarse enough, that we would not want to over-read the precise ratio between them: the qualitative story (T3 accumulates more transmission delay across more hops) is what the data supports; the exact "4×" figure is more fragile.

3.3 A3.3: Watching a Queue Fill

T3 (8.8.8.8) was pinged continuously for 5 minutes (300 packets) at a quiet hour (20-08-2026, 05:40) and again at a busy hour (22-08-2026, 15:11). Figure 3.3 plots RTT against sequence number for both.

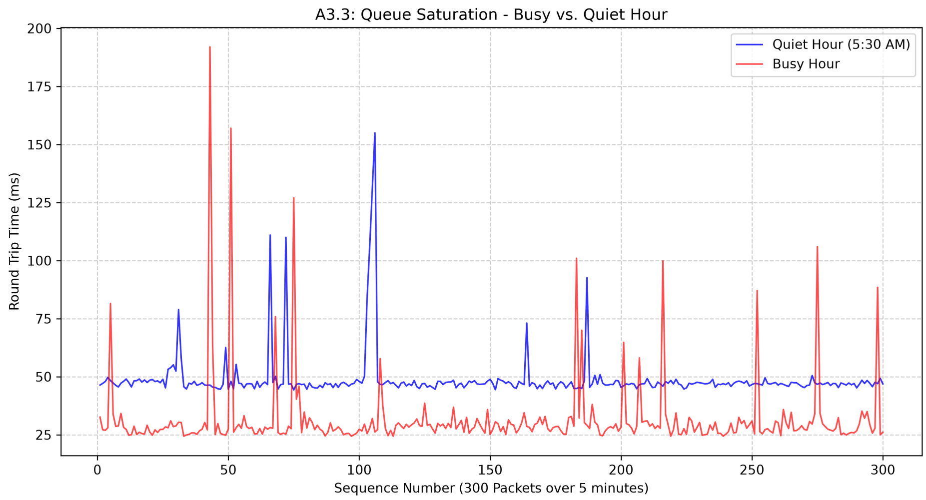


Figure 3.3: RTT vs. packet sequence number, 300 pings to T3, quiet hour (20-08-2026, 05:40) vs. busy hour (22-08-2026, 15:11). The plot legend rounds the quiet-hour start to 5:30 AM.

Which delay component moved, which didn't. The spikes in both traces (and there are noticeably more of them in the busy-hour trace) are queuing delay: propagation and transmission delay for a fixed-size packet on a fixed path do not vary probe-to-probe, so any variation has to be time spent sitting in a router buffer behind other traffic.

What we did not expect is that the two traces sit on different *baselines*: quiet hour settles around 48 ms, busy hour around 27 ms. Baseline (the floor propagation + transmission + fixed processing delay would produce with an empty queue) should not depend on time of day, so a 20 ms shift in the floor itself is not something queuing alone explains. The most likely account is that the two captures did not use the same path: something changed the route to 8.8.8.8 between the two windows, most plausibly Google's anycast routing handing us off to a different, physically closer edge for one of the two captures, though we cannot rule out a more mundane explanation on our end, such as the device having roamed to a different AP or uplink between captures. We do not have a way to confirm which, from ping data alone (that would need a traceroute run in each window, which we did not capture at the time), but whichever it is, it means this pair of captures is confounded for a pure "same path, different hour" queuing comparison, and the baseline shift itself is arguably the more interesting result of the two.

What the shape tells us. Looking past the baseline shift, the two traces do still differ in shape the way the assignment expects: the quiet-hour trace is flat with only a couple of isolated spikes (max $\sim$155 ms), consistent with a queue that is usually empty. The busy-hour trace is persistently spiky: frequent, sharp excursions up to $\sim$190 ms that return to baseline almost immediately rather than decaying gradually. That shape (sharp in, sharp out, rather than a sustained elevated floor) reads as bursty cross-traffic, short bursts that fill a buffer momentarily and drain quickly once the burst ends, rather than the queue being continuously close to full. So while we can't cleanly isolate "busy hour vs. quiet hour on the same path", what we can say is that the busy-hour capture, whatever path it took, shows a network under frequent transient load rather than sustained congestion.

Chapter 4

Evidence

The assignment requires 10–15 numbered evidence items, E-1 onward with no gaps, each stating who took it, when, where, the artefact itself, and which map element/table row/plot it supports.

E-1: photo

Who: Tejasvi Shrivastava (2024CS10582) When: 22-08-2026, 03:30 Where: NB-09, Satpura Hostel

Artefact: Figure 1.1, photo of the access point nearest Tejasvi's device, with handwritten placard (roll numbers, date, location).

Supports: Chapter 1, Section "Device and Access Point"; Figure 1.3.

E-2: photo

Who: Yashvardhan Singh Chaudhary (2024CS10300) When: 22-08-2026, 18:59 Where: NB-01, Satpura Hostel

Artefact: Figure 1.2, photo of the access point nearest Yashvardhan's device, with handwritten placard (roll numbers, date, location).

Supports: Chapter 1, Section "Device and Access Point"; Figure 1.3.

E-3: survey

Who: Tejasvi Shrivastava (2024CS10582) When: 22-08-2026, 05:27 (Satpura), with Lecture Hall and Library scans following the same session Where: Satpura Hostel, Lecture Hall, Library

Artefact: verbatim `nmcli -f BSSID,SSID,FREQ,CHAN,SIGNAL dev wifi` output, all three locations, untruncated (reproduced below; note the ∼5h27m gap the shell itself recorded between the Satpura and Lecture Hall scans).

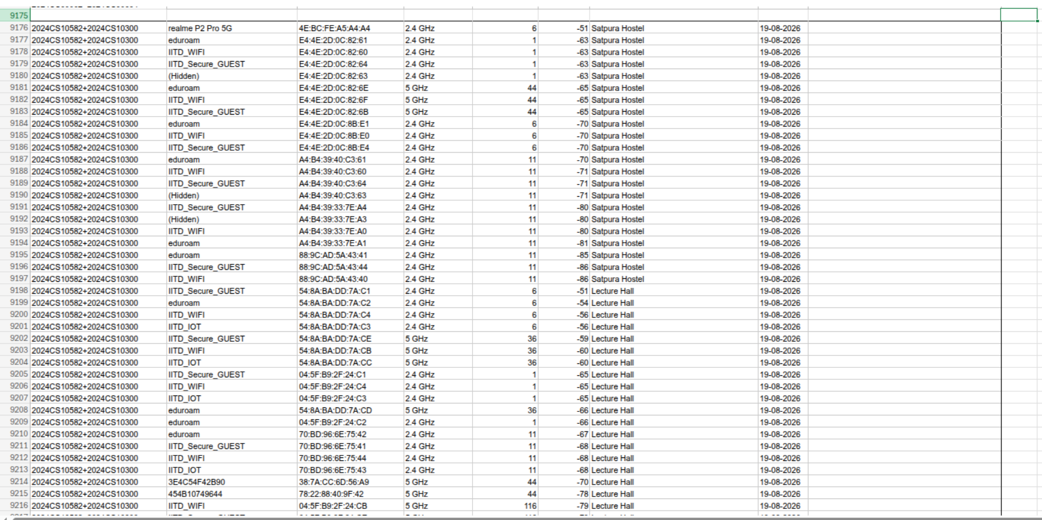
```
~
> nmcli -f BSSID,SSID,FREQ,CHAN,SIGNAL dev wifi
```

```
BSSID              SSID               FREQ      CHAN  SIGNAL
4E:BC:FE:A5:A4:A4  realme P2 Pro 5G   2437 MHz  6     97
E4:4E:2D:0C:82:61  eduroam            2412 MHz  1     74
E4:4E:2D:0C:82:60  IITD_WIFI          2412 MHz  1     74
 ... (22 BSSIDs total; full list in evidences/e2.txt)

~ took 5h27m2s
> nmcli -f BSSID,SSID,FREQ,CHAN,SIGNAL dev wifi
54:8A:BA:DD:7A:C1  IITD_Secure_GUEST  2437 MHz  6     97
 ... (21 BSSIDs total; full list in evidences/e2.txt)

~ took 6s
> nmcli -f BSSID,SSID,FREQ,CHAN,SIGNAL dev wifi
70:DF:2F:04:98:62  eduroam            2412 MHz  1     72
 ... (21 BSSIDs total; full list in evidences/e2.txt)
```

Supports: Chapter 2, Section A2.2 wireless survey table. (Note: the raw scan carries more BSSIDs per site than Table 2.2, since each physical AP was broadcasting the full MBSSID set on both bands here; Table 2.2 already reflects this in its "how many physical APs" count.) This same survey data was subsequently entered into the shared class map (Figure 4.1), so this item also supports Section A2.3's contribution count.



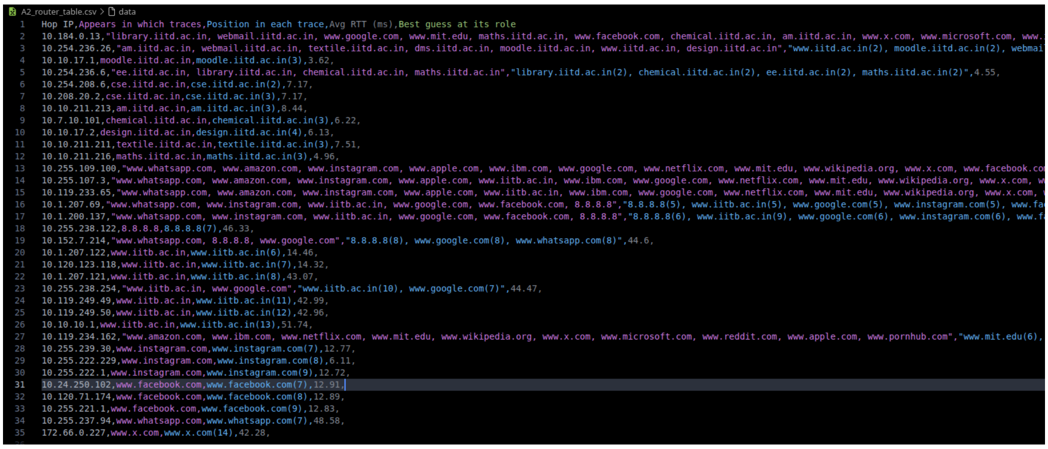
9175								
9176	2024CS10582+2024CS10300	realme P2 Pro 5G	4E:BC:FE:A5:A4:A4	2.4 GHz	6	-51	Satpura Hostel	19-08-2026
9177	2024CS10582+2024CS10300	eduroam	E4:4E:2D:0C:82:61	2.4 GHz	1	-63	Satpura Hostel	19-08-2026
9178	2024CS10582+2024CS10300	IITD_WIFI	E4:4E:2D:0C:82:60	2.4 GHz	1	-63	Satpura Hostel	19-08-2026
9179	2024CS10582+2024CS10300	IITD_Secure_GUEST	E4:4E:2D:0C:82:64	2.4 GHz	1	-63	Satpura Hostel	19-08-2026
9180	2024CS10582+2024CS10300	(Hidden)	E4:4E:2D:0C:82:63	2.4 GHz	1	-63	Satpura Hostel	19-08-2026
9181	2024CS10582+2024CS10300	eduroam	E4:4E:2D:0C:82:6E	5 GHz	44	-65	Satpura Hostel	19-08-2026
9182	2024CS10582+2024CS10300	IITD_WIFI	E4:4E:2D:0C:82:6F	5 GHz	44	-65	Satpura Hostel	19-08-2026
9183	2024CS10582+2024CS10300	IITD_Secure_GUEST	E4:4E:2D:0C:82:6B	5 GHz	44	-65	Satpura Hostel	19-08-2026
9184	2024CS10582+2024CS10300	eduroam	E4:4E:2D:0C:8B:E1	2.4 GHz	6	-70	Satpura Hostel	19-08-2026
9185	2024CS10582+2024CS10300	IITD_WIFI	E4:4E:2D:0C:8B:E0	2.4 GHz	6	-70	Satpura Hostel	19-08-2026
9186	2024CS10582+2024CS10300	IITD_Secure_GUEST	E4:4E:2D:0C:8B:E4	2.4 GHz	6	-70	Satpura Hostel	19-08-2026
9187	2024CS10582+2024CS10300	eduroam	A4:B4:39:40:C3:61	2.4 GHz	11	-70	Satpura Hostel	19-08-2026
9188	2024CS10582+2024CS10300	IITD_WIFI	A4:B4:39:40:C3:60	2.4 GHz	11	-71	Satpura Hostel	19-08-2026
9189	2024CS10582+2024CS10300	IITD_Secure_GUEST	A4:B4:39:40:C3:64	2.4 GHz	11	-71	Satpura Hostel	19-08-2026
9190	2024CS10582+2024CS10300	(Hidden)	A4:B4:39:40:C3:63	2.4 GHz	11	-71	Satpura Hostel	19-08-2026
9191	2024CS10582+2024CS10300	IITD_Secure_GUEST	A4:B4:39:33:7E:A4	2.4 GHz	11	-80	Satpura Hostel	19-08-2026
9192	2024CS10582+2024CS10300	(Hidden)	A4:B4:39:33:7E:A3	2.4 GHz	11	-80	Satpura Hostel	19-08-2026
9193	2024CS10582+2024CS10300	IITD_WIFI	A4:B4:39:33:7E:A0	2.4 GHz	11	-80	Satpura Hostel	19-08-2026
9194	2024CS10582+2024CS10300	eduroam	A4:B4:39:33:7E:A1	2.4 GHz	11	-81	Satpura Hostel	19-08-2026
9195	2024CS10582+2024CS10300	eduroam	88:9C:AD:5A:43:41	2.4 GHz	11	-85	Satpura Hostel	19-08-2026
9196	2024CS10582+2024CS10300	IITD_Secure_GUEST	88:9C:AD:5A:43:44	2.4 GHz	11	-86	Satpura Hostel	19-08-2026
9197	2024CS10582+2024CS10300	IITD_WIFI	88:9C:AD:5A:43:40	2.4 GHz	11	-86	Satpura Hostel	19-08-2026
9198	2024CS10582+2024CS10300	IITD_Secure_GUEST	54:8A:BA:DD:7A:C1	2.4 GHz	6	-51	Lecture Hall	19-08-2026
9199	2024CS10582+2024CS10300	eduroam	54:8A:BA:DD:7A:C2	2.4 GHz	6	-54	Lecture Hall	19-08-2026
9200	2024CS10582+2024CS10300	IITD_WIFI	54:8A:BA:DD:7A:C4	2.4 GHz	6	-56	Lecture Hall	19-08-2026
9201	2024CS10582+2024CS10300	IITD_IOT	54:8A:BA:DD:7A:C3	2.4 GHz	6	-56	Lecture Hall	19-08-2026
9202	2024CS10582+2024CS10300	IITD_Secure_GUEST	54:8A:BA:DD:7A:CE	5 GHz	36	-59	Lecture Hall	19-08-2026
9203	2024CS10582+2024CS10300	IITD_WIFI	54:8A:BA:DD:7A:CB	5 GHz	36	-60	Lecture Hall	19-08-2026
9204	2024CS10582+2024CS10300	IITD_IOT	54:8A:BA:DD:7A:CC	5 GHz	36	-60	Lecture Hall	19-08-2026
9205	2024CS10582+2024CS10300	IITD_Secure_GUEST	04:5F:B9:2F:24:C1	2.4 GHz	1	-65	Lecture Hall	19-08-2026
9206	2024CS10582+2024CS10300	IITD_WIFI	04:5F:B9:2F:24:C4	2.4 GHz	1	-65	Lecture Hall	19-08-2026
9207	2024CS10582+2024CS10300	IITD_IOT	04:5F:B9:2F:24:C3	2.4 GHz	1	-65	Lecture Hall	19-08-2026
9208	2024CS10582+2024CS10300	eduroam	54:8A:BA:DD:7A:CD	5 GHz	36	-66	Lecture Hall	19-08-2026
9209	2024CS10582+2024CS10300	eduroam	04:5F:B9:2F:24:C2	2.4 GHz	1	-66	Lecture Hall	19-08-2026
9210	2024CS10582+2024CS10300	eduroam	70:BD:96:6E:75:42	2.4 GHz	11	-67	Lecture Hall	19-08-2026
9211	2024CS10582+2024CS10300	IITD_Secure_GUEST	70:BD:96:6E:75:41	2.4 GHz	11	-68	Lecture Hall	19-08-2026
9212	2024CS10582+2024CS10300	IITD_WIFI	70:BD:96:6E:75:44	2.4 GHz	11	-68	Lecture Hall	19-08-2026
9213	2024CS10582+2024CS10300	IITD_IOT	70:BD:96:6E:75:43	2.4 GHz	11	-68	Lecture Hall	19-08-2026
9214	2024CS10582+2024CS10300	3E4C54F42B90	38:7A:CC:6D:56:A9	5 GHz	44	-70	Lecture Hall	19-08-2026
9215	2024CS10582+2024CS10300	454B10749644	78:22:88:40:9F:42	5 GHz	44	-78	Lecture Hall	19-08-2026
9216	2024CS10582+2024CS10300	IITD_WIFI	04:5F:B9:2F:24:CB	5 GHz	116	-79	Lecture Hall	19-08-2026

Figure 4.1: Our wireless-survey rows (Evidence E-3) entered into `COL334_A1_Campus_Map.xlsx`, spanning all three surveyed locations.

E-4: spreadsheet

Who: Tejasvi Shrivastava (2024CS10582) When: 19-08-2026, 19:30 Where: NB-09, Satpura Hostel

Artefact: Figure 4.2: `A2_router_table.csv`, the raw per-trace hop data underlying Table 2.1.

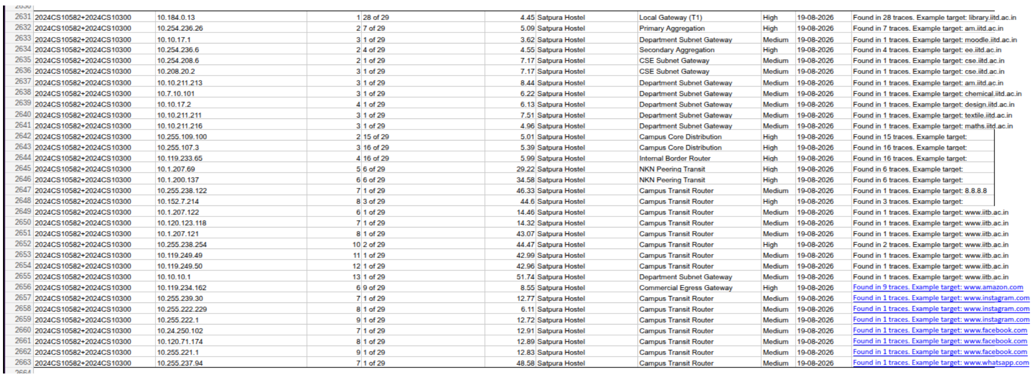


A2_router_table.csv > data

```
Hop IP,Appears in which traces,Position in each trace,Avg RTT (ms),Best guess at its role
10.184.0.13,"library.iitd.ac.in, webmail.iitd.ac.in, www.google.com, www.mit.edu, maths.iitd.ac.in, www.facebook.com, chemical.iitd.ac.in, am.iitd.ac.in, www.x.com, www.microsoft.com, www.
10.254.236.26,"am.iitd.ac.in, webmail.iitd.ac.in, textile.iitd.ac.in, dms.iitd.ac.in, moodle.iitd.ac.in, www.iitd.ac.in, design.iitd.ac.in","www.iitd.ac.in(2), moodle.iitd.ac.in(2), webmai
10.10.17.1,moodle.iitd.ac.in,moodle.iitd.ac.in(3),3.62,
10.254.236.6,"ee.iitd.ac.in, library.iitd.ac.in, chemical.iitd.ac.in, maths.iitd.ac.in","library.iitd.ac.in(2), chemical.iitd.ac.in(2), ee.iitd.ac.in(2), maths.iitd.ac.in(2)",4.55,
10.254.208.6,cse.iitd.ac.in,cse.iitd.ac.in(2),7.17,
10.208.20.2,cse.iitd.ac.in,cse.iitd.ac.in(3),7.17,
10.10.211.213,am.iitd.ac.in,am.iitd.ac.in(3),8.44,
10.7.10.101,chemical.iitd.ac.in,chemical.iitd.ac.in(3),6.22,
10.10.17.2,design.iitd.ac.in,design.iitd.ac.in(4),6.13,
10.10.211.211,textile.iitd.ac.in,textile.iitd.ac.in(3),7.51,
10.10.211.216,maths.iitd.ac.in,maths.iitd.ac.in(3),4.96,
10.255.109.100,"www.whatsapp.com, www.amazon.com, www.instagram.com, www.apple.com, www.ibm.com, www.google.com, www.netflix.com, www.mit.edu, www.wikipedia.org, www.x.com, www.facebook.co
10.255.107.3,"www.whatsapp.com, www.amazon.com, www.instagram.com, www.apple.com, www.iitb.ac.in, www.ibm.com, www.google.com, www.netflix.com, www.mit.edu, www.wikipedia.org, www.x.com, w
10.119.233.65,"www.whatsapp.com, www.amazon.com, www.instagram.com, www.apple.com, www.iitb.ac.in, www.ibm.com, www.google.com, www.netflix.com, www.mit.edu, www.wikipedia.org, www.x.com, 
10.1.207.69,"www.whatsapp.com, www.instagram.com, www.iitb.ac.in, www.google.com, www.facebook.com, 8.8.8.8","8.8.8.8(5), www.iitb.ac.in(5), www.google.com(5), www.instagram.com(5), www.fa
10.1.200.137,"www.whatsapp.com, www.instagram.com, www.iitb.ac.in, www.google.com, www.facebook.com, 8.8.8.8","8.8.8.8(6), www.iitb.ac.in(9), www.google.com(6), www.instagram.com(6), www.f
10.255.238.122,8.8.8.8,8.8.8.8(7),46.33,
10.152.7.214,"www.whatsapp.com, 8.8.8.8, www.google.com","8.8.8.8(8), www.google.com(8), www.whatsapp.com(8)",44.6,
10.1.207.122,www.iitb.ac.in,www.iitb.ac.in(6),14.46,
10.120.123.118,www.iitb.ac.in,www.iitb.ac.in(7),14.32,
10.1.207.121,www.iitb.ac.in,www.iitb.ac.in(8),43.07,
10.255.238.254,"www.iitb.ac.in, www.google.com","www.iitb.ac.in(10), www.google.com(7)",44.47,
10.119.249.49,www.iitb.ac.in,www.iitb.ac.in(11),42.99,
10.119.249.50,www.iitb.ac.in,www.iitb.ac.in(12),42.96,
10.10.10.1,www.iitb.ac.in,www.iitb.ac.in(13),51.74,
10.119.234.162,"www.amazon.com, www.ibm.com, www.netflix.com, www.mit.edu, www.wikipedia.org, www.x.com, www.microsoft.com, www.reddit.com, www.apple.com, www.pornhub.com","www.mit.edu(6),
10.255.239.30,www.instagram.com,www.instagram.com(7),12.77,
10.255.222.229,www.instagram.com,www.instagram.com(8),6.11,
10.255.222.1,www.instagram.com,www.instagram.com(9),12.72,
10.24.250.102,www.facebook.com,www.facebook.com(7),12.91,
10.120.71.174,www.facebook.com,www.facebook.com(8),12.89,
10.255.221.1,www.facebook.com,www.facebook.com(9),12.83,
10.255.237.94,www.whatsapp.com,www.whatsapp.com(7),48.58,
172.66.0.227,www.x.com,www.x.com(14),42.28,
```

Figure 4.2: Raw A2_router_table.csv data underlying Table 2.1. More hops appear here than in the table; the extras are single-appearance transit nodes on paths to sites not otherwise discussed (e.g. facebook.com, x.com) and were left out of Table 2.1 to keep it to the destinations this report actually analyses.

This same router data was entered into the shared class map as our A2.3 contribution, split between internal (Figure 4.3) and external/CDN (Figure 4.4) hops:



2631	2024CS10582+2024CS10300	10.184.0.13	1	28 of 29	4.45	Satpura Hostel	Local Gateway (T1)	High	19-08-2026	Found in 28 traces. Example target: library.iitd.ac.in
2632	2024CS10582+2024CS10300	10.254.236.26	2	7 of 29	5.09	Satpura Hostel	Primary Aggregation	High	19-08-2026	Found in 7 traces. Example target: am.iitd.ac.in
2633	2024CS10582+2024CS10300	10.10.17.1	3	1 of 29	3.62	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: moodle.iitd.ac.in
2634	2024CS10582+2024CS10300	10.254.236.6	2	4 of 29	4.55	Satpura Hostel	Secondary Aggregation	High	19-08-2026	Found in 4 traces. Example target: ee.iitd.ac.in
2635	2024CS10582+2024CS10300	10.254.208.6	2	1 of 29	7.17	Satpura Hostel	CSE Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: cse.iitd.ac.in
2636	2024CS10582+2024CS10300	10.208.20.2	3	1 of 29	7.17	Satpura Hostel	CSE Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: cse.iitd.ac.in
2637	2024CS10582+2024CS10300	10.10.211.213	3	1 of 29	8.44	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: am.iitd.ac.in
2638	2024CS10582+2024CS10300	10.7.10.101	3	1 of 29	6.22	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: chemical.iitd.ac.in
2639	2024CS10582+2024CS10300	10.10.17.2	4	1 of 29	6.13	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: design.iitd.ac.in
2640	2024CS10582+2024CS10300	10.10.211.211	3	1 of 29	7.51	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: textile.iitd.ac.in
2641	2024CS10582+2024CS10300	10.10.211.216	3	1 of 29	4.96	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: maths.iitd.ac.in
2642	2024CS10582+2024CS10300	10.255.109.100	2	15 of 29	5.01	Satpura Hostel	Campus Core Distribution	High	19-08-2026	Found in 15 traces. Example target:
2643	2024CS10582+2024CS10300	10.255.107.3	3	16 of 29	5.39	Satpura Hostel	Campus Core Distribution	High	19-08-2026	Found in 16 traces. Example target:
2644	2024CS10582+2024CS10300	10.119.233.65	4	16 of 29	5.99	Satpura Hostel	Internal Border Router	High	19-08-2026	Found in 16 traces. Example target:
2645	2024CS10582+2024CS10300	10.1.207.69	5	6 of 29	29.22	Satpura Hostel	NKN Peering Transit	High	19-08-2026	Found in 6 traces. Example target:
2646	2024CS10582+2024CS10300	10.1.200.137	6	6 of 29	34.58	Satpura Hostel	NKN Peering Transit	High	19-08-2026	Found in 6 traces. Example target:
2647	2024CS10582+2024CS10300	10.255.238.122	7	1 of 29	46.33	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: 8.8.8.8
2648	2024CS10582+2024CS10300	10.152.7.214	8	3 of 29	44.6	Satpura Hostel	Campus Transit Router	High	19-08-2026	Found in 3 traces. Example target:
2649	2024CS10582+2024CS10300	10.1.207.122	6	1 of 29	14.46	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2650	2024CS10582+2024CS10300	10.120.123.118	7	1 of 29	14.32	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2651	2024CS10582+2024CS10300	10.1.207.121	8	1 of 29	43.07	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2652	2024CS10582+2024CS10300	10.255.238.254	10	2 of 29	44.47	Satpura Hostel	Campus Transit Router	High	19-08-2026	Found in 2 traces. Example target: www.iitb.ac.in
2653	2024CS10582+2024CS10300	10.119.249.49	11	1 of 29	42.99	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2654	2024CS10582+2024CS10300	10.119.249.50	12	1 of 29	42.96	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2655	2024CS10582+2024CS10300	10.10.10.1	13	1 of 29	51.74	Satpura Hostel	Department Subnet Gateway	Medium	19-08-2026	Found in 1 traces. Example target: www.iitb.ac.in
2656	2024CS10582+2024CS10300	10.119.234.162	6	9 of 29	8.55	Satpura Hostel	Commercial Egress Gateway	High	19-08-2026	Found in 9 traces. Example target: www.amazon.com
2657	2024CS10582+2024CS10300	10.255.239.30	7	1 of 29	12.77	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2658	2024CS10582+2024CS10300	10.255.222.229	8	1 of 29	6.11	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2659	2024CS10582+2024CS10300	10.255.222.1	9	1 of 29	12.72	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2660	2024CS10582+2024CS10300	10.24.250.102	7	1 of 29	12.91	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.facebook.com
2661	2024CS10582+2024CS10300	10.120.71.174	8	1 of 29	12.89	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.facebook.com
2662	2024CS10582+2024CS10300	10.255.221.1	9	1 of 29	12.83	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.facebook.com
2663	2024CS10582+2024CS10300	10.255.237.94	7	1 of 29	48.58	Satpura Hostel	Campus Transit Router	Medium	19-08-2026	Found in 1 traces. Example target: www.whatsapp.com

Figure 4.3: Our internal-hop rows in COL334_A1_Campus_Map.xlsx.

2656	2024CS10582+2024CS10300	8.8.8.8	11	1 of 28	63.75	Satpura Hostel	Google DNS Server (Destination)	Medium	19-08-2026	Found in 1 traces. Target: 8.8.8.8
2657	2024CS10582+2024CS10300	59.144.234.93	7	5 of 28	22.86	Satpura Hostel	CDN Edge / International Transit	High	19-08-2026	Found in 5 traces. Example target: www.microsoft.com
2658	2024CS10582+2024CS10300	116.119.42.218	2	2 of 28	88.23	Satpura Hostel	CDN Edge / International Transit	High	19-08-2026	Found in 2 traces. Example target: www.mit.edu
2659	2024CS10582+2024CS10300	23.54.79.22	10	1 of 28	88.68	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.mit.edu
2660	2024CS10582+2024CS10300	23.54.79.11	11	1 of 28	87.55	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.mit.edu
2661	2024CS10582+2024CS10300	23.54.79.4	12	1 of 28	75.47	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.mit.edu
2662	2024CS10582+2024CS10300	23.54.118.193	13	1 of 28	86.31	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.mit.edu
2663	2024CS10582+2024CS10300	72.14.204.62	9	1 of 28	60.05	Satpura Hostel	External Transit / ISP Node	Medium	19-08-2026	Found in 1 traces. Example target: www.google.com
2664	2024CS10582+2024CS10300	142.251.151.119	10	1 of 28	71.38	Satpura Hostel	External Transit / ISP Node	Medium	19-08-2026	Found in 1 traces. Example target: www.google.com
2665	2024CS10582+2024CS10300	157.240.68.238	9	1 of 28	58.32	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2666	2024CS10582+2024CS10300	129.134.54.128	10	1 of 28	57.13	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2667	2024CS10582+2024CS10300	129.134.84.43	11	1 of 28	72.39	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.instagram.com
2668	2024CS10582+2024CS10300	157.240.65.230	9	2 of 28	131.77	Satpura Hostel	Meta Peering / Transit Edge	High	19-08-2026	Found in 2 traces. Example target: www.whatsapp.com
2669	2024CS10582+2024CS10300	129.134.47.42	10	1 of 28	120.24	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.facebook.com
2670	2024CS10582+2024CS10300	129.134.89.205	11	1 of 28	117.06	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.facebook.com
2671	2024CS10582+2024CS10300	116.119.121.117	8	1 of 28	75.01	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.whatsapp.com
2672	2024CS10582+2024CS10300	129.134.46.110	10	1 of 28	130.58	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.whatsapp.com
2673	2024CS10582+2024CS10300	129.134.95.164	11	1 of 28	117.63	Satpura Hostel	Meta Peering / Transit Edge	Medium	19-08-2026	Found in 1 traces. Example target: www.whatsapp.com
2674	2024CS10582+2024CS10300	162.158.226.122	12	1 of 28	74.14	Satpura Hostel	External Transit / ISP Node	Medium	19-08-2026	Found in 1 traces. Example target: www.x.com
2675	2024CS10582+2024CS10300	162.158.226.71	13	1 of 28	62.31	Satpura Hostel	External Transit / ISP Node	Medium	19-08-2026	Found in 1 traces. Example target: www.x.com
2676	2024CS10582+2024CS10300	162.159.140.229	14	1 of 28	62.28	Satpura Hostel	External Transit / ISP Node	Medium	19-08-2026	Found in 1 traces. Example target: www.x.com
2677	2024CS10582+2024CS10300	136.232.148.177	7	2 of 28	16.54	Satpura Hostel	External Transit / ISP Node	High	19-08-2026	Found in 2 traces. Example target: www.amazon.com
2678	2024CS10582+2024CS10300	14.143.159.145	7	1 of 28	16.43	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2679	2024CS10582+2024CS10300	14.143.59.206	10	1 of 28	67.51	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2680	2024CS10582+2024CS10300	104.70.119.38	11	1 of 28	70.82	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2681	2024CS10582+2024CS10300	23.65.190.73	12	1 of 28	89.18	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2682	2024CS10582+2024CS10300	104.70.116.37	13	1 of 28	79.22	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2683	2024CS10582+2024CS10300	104.70.116.25	14	1 of 28	73.56	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.apple.com
2684	2024CS10582+2024CS10300	116.119.119.198	8	1 of 28	89.61	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.microsoft.com
2685	2024CS10582+2024CS10300	23.54.79.23	10	1 of 28	87.36	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.microsoft.com
2686	2024CS10582+2024CS10300	23.54.79.19	11	1 of 28	75.41	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.microsoft.com
2687	2024CS10582+2024CS10300	23.54.79.42	12	1 of 28	87.28	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.microsoft.com
2688	2024CS10582+2024CS10300	23.193.164.161	13	1 of 28	75.33	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.microsoft.com
2689	2024CS10582+2024CS10300	23.56.138.79	9	1 of 28	86.8	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.ibm.com
2690	2024CS10582+2024CS10300	23.56.138.19	10	1 of 28	122.56	Satpura Hostel	CDN Edge / International Transit	Medium	19-08-2026	Found in 1 traces. Example target: www.ibm.com

Figure 4.4: Our external transit and CDN-edge rows in `COL334_A1_Campus_Map.xlsx`.

Supports: Chapter 2, Table 2.1, Figure 2.1, and Section A2.3 (contribution count).

E-5: command output

Who: Tejasvi Shrivastava (2024CS10582) When: 22-08-2026, 16:15 Where: NB-09, Satpura Hostel

Artefact: `ip addr` (confirming interface `wlo1`) and `traceroute -n www.codechef.com`, verbatim:

```
> traceroute -n www.codechef.com
traceroute to www.codechef.com (172.67.75.52), 30 hops max, 60 byte packets
 1  10.184.0.13     2.759 ms  2.676 ms  2.644 ms
 2  10.255.109.100  2.527 ms  2.498 ms  2.472 ms
 3  10.255.107.3    3.784 ms  3.759 ms  4.793 ms
 4  10.119.233.65   3.703 ms  3.679 ms  3.652 ms
 5  * * *
 6  10.119.234.162  4.620 ms  4.465 ms  4.341 ms
 7  14.143.159.145  15.750 ms * 26.497 ms
 8  * * *
 9  * 121.240.255.30  6.970 ms *
10  104.23.231.5 8.611 ms * 104.23.231.30  16.686 ms
11  172.67.75.52    16.622 ms  4.546 ms  14.703 ms
```

(This is a separate run from the one quoted in Section 1.3; hop sequence and roles are identical, RTTs differ run-to-run as expected.)

Supports: Chapter 1, Section "Finding T1"; Figure 1.3 (shared-gateway note).

E-6: command output

Who: Yashvardhan Singh Chaudhary (2024CS10300) When: 22-08-2026, 16:25 Where: NB-01, Satpura Hostel

Artefact: `ipconfig` and `tracert -d www.codechef.com`, run on Windows, verbatim:

```
Wireless LAN adapter Wi-Fi:
```

```
   Connection-specific DNS Suffix  . : iitd.ac.in
   IPv4 Address. . . . . . . . . . . : 10.184.20.91
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Default Gateway . . . . . . . . . : 10.184.0.1

> tracert -d www.codechef.com
Tracing route to www.codechef.com [172.67.75.52]
 1     6 ms     4 ms     2 ms  10.184.0.13
 2     8 ms     8 ms    13 ms  10.255.109.100
 3    15 ms    15 ms    29 ms  10.255.107.3
 4     6 ms     3 ms     2 ms  10.119.233.65
 5     *        *        *     Request timed out.
 6     5 ms     4 ms     5 ms  10.119.234.162
 7    17 ms    16 ms    18 ms  14.143.159.145
 8     *        *        *     Request timed out.
 9    34 ms     *       47 ms  121.240.255.30
10     *       34 ms    21 ms  104.23.231.30
11     7 ms     7 ms     9 ms  172.67.75.52
Trace complete.
```

This independently confirms two things from a second device on a second OS: the same **10.184.0.1** DHCP gateway / **10.184.0.13** first-hop discrepancy noted in Section 1.2, and the identical hop-5/hop-8 timeout pattern discussed in the diagram's analytical notes: since both teammates see the same two hops go silent independently, a probe-specific fluke on one device is less likely than a deliberate no-reply configuration on those routers.

Supports: Chapter 1, Section "Finding T1"; Figure 1.3 (shared-gateway note).

E-7: command output / lookup

Who: Tejasvi Shrivastava (2024CS10582) When: 19-08-2026, 19:35 Where: NB-09, Satpura Hostel

Artefact: `whois` lookups for the four public hops in Table 1.1, verbatim:

```
> whois 14.143.159.145 | grep -iE "orgname|netname|descr"
netname:        TATACOMM-IN
descr:          Internet Service Provider
descr:          TATA Communications formerly VSNL is Leading ISP,
descr:          Data and Voice Carrier in India
descr:          Tata Communications Limited

> whois 121.240.255.30 | grep -iE "orgname|netname|descr"
netname:        TATACOMM-IN
descr:          Route for VSNL

> whois 104.23.231.30 | grep -iE "orgname|netname|descr"
NetName:        CLOUDFLARENET
```

```
OrgName:        Cloudflare, Inc.

> whois 172.67.75.52 | grep -iE "orgname|netname|descr"
NetName:        CLOUDFLARENET
OrgName:        Cloudflare, Inc.
```

Supports: Chapter 1, Section A1.3, Table 1.1.

E-8: log

Who: Tejasvi Shrivastava (2024CS10582) When: quiet hour, 20-08-2026, 05:40 Where: NB-09, Satpura Hostel

Artefact: 300-packet continuous ping to T3, quiet hour, first/last 10 lines:

```
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=46.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=47.1 ms
 ... (280 lines omitted; full log in evidences/e7.txt) ...
64 bytes from 8.8.8.8: icmp_seq=299 icmp_seq=299 ttl=112 time=49.3 ms
64 bytes from 8.8.8.8: icmp_seq=300 ttl=112 time=46.9 ms

--- 8.8.8.8 ping statistics ---
300 packets transmitted, 300 received, 0% packet loss, time 299257ms
rtt min/avg/max/mdev = 44.269/48.832/154.554/10.852 ms
```

Supports: Chapter 3, Figure 3.3.

E-9: log

Who: Tejasvi Shrivastava (2024CS10582) When: busy hour, 22-08-2026, 15:11 Where: NB-09, Satpura Hostel

Artefact: 300-packet continuous ping to T3, busy hour, first/last 10 lines:

```
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=32.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=27.2 ms
 ... (280 lines omitted; full log in evidences/e8.txt) ...
64 bytes from 8.8.8.8: icmp_seq=299 ttl=114 time=25.0 ms
64 bytes from 8.8.8.8: icmp_seq=300 ttl=114 time=26.1 ms

--- 8.8.8.8 ping statistics ---
300 packets transmitted, 300 received, 0% packet loss, time 299353ms
rtt min/avg/max/mdev = 24.356/31.706/192.284/17.041 ms
```

Supports: Chapter 3, Figure 3.3.

E-10: command output

Who: Tejasvi Shrivastava (2024CS10582) When: 19-08-2026, 19:35 Where: NB-09, Satpura Hostel

Artefact: 50-packet ping summaries for T1 and T2:

```
> ping -c 50 -s 64 10.184.0.13
--- 10.184.0.13 ping statistics ---
50 packets transmitted, 50 received, 0% packet loss, time 49072ms
rtt min/avg/max/mdev = 3.904/8.891/96.290/12.787 ms

> ping -c 50 -s 64 www.iitd.ac.in
--- www.iitd.ac.in ping statistics ---
50 packets transmitted, 50 received, 0% packet loss, time 49078ms
rtt min/avg/max/mdev = 3.315/7.489/14.110/2.716 ms
```

Supports: Chapter 3, Table 3.1, Figure 3.1.

E-11: command output

Who: Tejasvi Shrivastava (2024CS10582) When: 20-08-2026, 04:50 Where: NB-09, Satpura Hostel

Artefact: 50-packet ping summaries for T3 and T5 (note the minor packet loss on both, 2% and 4% respectively, not previously called out in Chapter 3):

```
> ping -c 50 -s 64 8.8.8.8
--- 8.8.8.8 ping statistics ---
50 packets transmitted, 49 received, 2% packet loss, time 49119ms
rtt min/avg/max/mdev = 42.824/49.859/119.830/11.032 ms

> ping -c 50 -s 64 www.mit.edu
--- www.mit.edu ping statistics ---
50 packets transmitted, 48 received, 4% packet loss, time 49175ms
rtt min/avg/max/mdev = 35.596/51.145/247.982/36.326 ms
```

Supports: Chapter 3, Table 3.1, Figure 3.1.

E-12: command output

Who: Tejasvi Shrivastava (2024CS10582) When: 20-08-2026, 04:50 Where: NB-09, Satpura Hostel

Artefact: 100% packet loss for T4:

```
> ping -c 50 -s 64 www.iitb.ac.in
PING www.iitb.ac.in (103.21.124.133) 64(92) bytes of data.
--- www.iitb.ac.in ping statistics ---
50 packets transmitted, 0 received, 100% packet loss, time 50162ms
```

Supports: Chapter 3, Table 3.1, Section A3.1 firewall-drop discussion.

E-13: command output

Who: Tejasvi Shrivastava (2024CS10582) When: 19-08-2026, 20:00 Where: NB-09, Satpura Hostel

Artefact: packet-size sweep summaries for T1 (64/256/512/1024/1400 bytes):

```
--- 10.184.0.13 ping statistics --- (size 64)
rtt min/avg/max/mdev = 3.186/10.550/110.213/20.142 ms
--- 10.184.0.13 ping statistics --- (size 256)
rtt min/avg/max/mdev = 2.838/5.688/27.777/4.310 ms
--- 10.184.0.13 ping statistics --- (size 512)
rtt min/avg/max/mdev = 2.891/6.757/44.268/7.115 ms
--- 10.184.0.13 ping statistics --- (size 1024)
rtt min/avg/max/mdev = 3.043/6.273/20.171/3.219 ms
--- 10.184.0.13 ping statistics --- (size 1400)
rtt min/avg/max/mdev = 3.202/6.025/11.310/1.969 ms
```

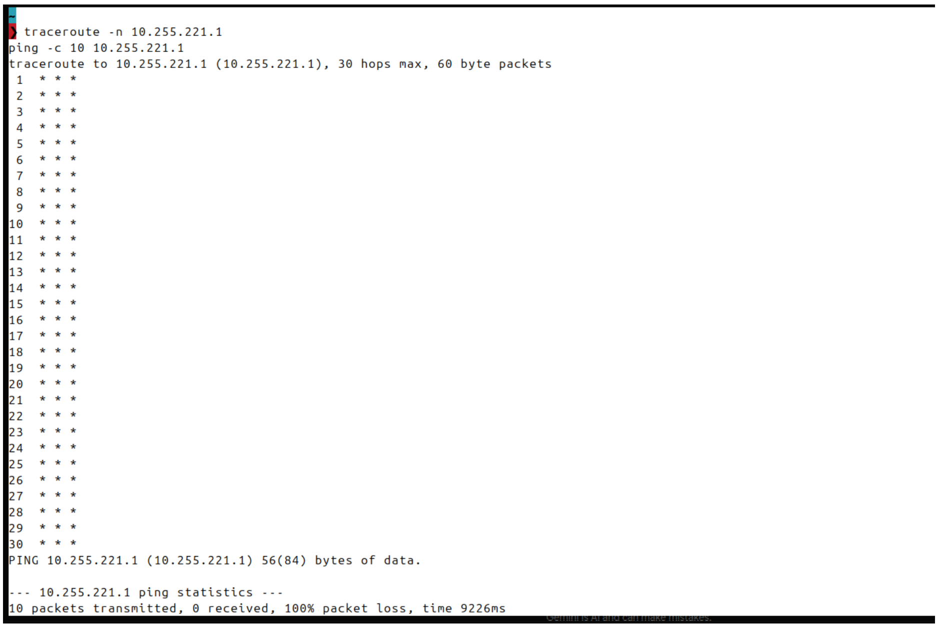
Supports: Chapter 3, Figure 3.2, T1 regression.

E-14: command output
Who: Tejasvi Shrivastava (2024CS10582) When: 19-08-2026, 20:15 Where: NB-09, Satpura Hostel
Artefact: packet-size sweep summaries for T3 (64/256/512/1024/1400 bytes):

```
--- 8.8.8.8 ping statistics --- (size 64)
rtt min/avg/max/mdev = 44.637/46.964/52.083/1.430 ms
--- 8.8.8.8 ping statistics --- (size 256)
rtt min/avg/max/mdev = 44.587/52.202/183.043/24.330 ms
--- 8.8.8.8 ping statistics --- (size 512)
rtt min/avg/max/mdev = 44.733/50.704/83.341/7.764 ms
--- 8.8.8.8 ping statistics --- (size 1024)
rtt min/avg/max/mdev = 44.709/49.003/84.001/6.643 ms
--- 8.8.8.8 ping statistics --- (size 1400)
rtt min/avg/max/mdev = 45.368/47.889/52.012/1.350 ms
```

Supports: Chapter 3, Figure 3.2, T3 regression.

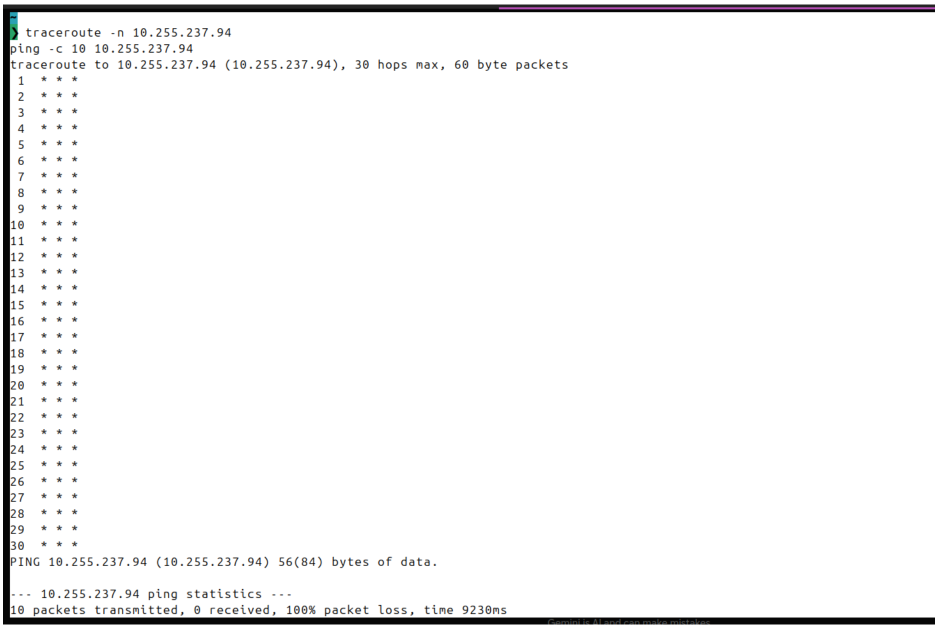
E-15: command output
Who: Tejasvi Shrivastava (2024CS10582) When: 22-08-2026, 20:43 Where: NB-09, Satpura Hostel
Artefact: independent `traceroute -n` + `ping -c 10` verification of two rows submitted by team 2024CS50261+2024CS51073, both returning full timeouts:



```
❯ traceroute -n 10.255.221.1
ping -c 10 10.255.221.1
traceroute to 10.255.221.1 (10.255.221.1), 30 hops max, 60 byte packets
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
PING 10.255.221.1 (10.255.221.1) 56(84) bytes of data.

--- 10.255.221.1 ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9226ms
```

Figure 4.5: Row 1092 (10.255.221.1, claimed hop 11): 30/30 traceroute hops and 10/10 pings timed out. Result: couldn't reach.



```
❯ traceroute -n 10.255.237.94
ping -c 10 10.255.237.94
traceroute to 10.255.237.94 (10.255.237.94), 30 hops max, 60 byte packets
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
PING 10.255.237.94 (10.255.237.94) 56(84) bytes of data.

--- 10.255.237.94 ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9230ms
```

Figure 4.6: Row 1096 (10.255.237.94, claimed hop 7): 30/30 traceroute hops and 10/10 pings timed out. Result: couldn't reach.

Supports: Chapter 2, Section A2.3 (verification).

Chapter 5

Three Findings

F-1: Diagnostic Blackholes in Transit
Saw: Hops 5 and 8 in the traceroute to `www.codechef.com` returned 100% timeouts (`* * *`) on both teammates' independent runs.
Evidence: E-5, E-6
Think: We conclude that these specific transit routers are configured not to generate ICMP time-exceeded replies, obscuring internal infrastructure from standard traceroute-based mapping. One other explanation we couldn't rule out is that these routers have a rate-limited ICMP control plane that happened to be saturated by background traffic on both our probes, dropping our specific packets rather than enforcing a hard no-reply policy, though the fact that *both* teammates saw exactly the same two hops go silent, on different days and devices, makes coincidental rate-limiting the less likely of the two.
Fix: Enable ICMP time-exceeded (Type 11, Code 0) generation on these internal routers, with a conservative rate limit (e.g. 50 pps) so legitimate diagnostics and PMTUD keep working without opening a full information-disclosure path.
Check: Re-run `traceroute -n www.codechef.com`; hops 5 and 8 should return valid IPs and RTTs consistent with their neighbours, instead of asterisks.

F-2: First-Hop Wi-Fi Delay Floor
Saw: The minimum RTT to the immediate local gateway (10.184.0.13) across 50 probes was 3.904 ms, not the sub-millisecond floor a single wired hop would suggest.
Evidence: E-10
Think: We conclude this reflects the Wi-Fi RF environment in the hostel: CSMA/CA contention and backoff on a busy 2.4/5 GHz band, rather than anything downstream of the gateway itself, consistent with our reading of this floor in Section 3.1. One other explanation we couldn't rule out is that the access point's CPU is under load from

managing a large number of simultaneous MBSSID client associations, adding a small software-level queuing delay before a packet even reaches the routing fabric: a device-side bottleneck rather than a wireless-medium one.

Fix: Add access points in the hostel corridors and reduce transmit power on existing ones to shrink cell size and cut co-channel contention, rather than relying on the current handful of APs to cover the whole floor.

Check: Repeat `ping -c 50 10.184.0.13` from the same location; if the RF explanation is right, the minimum RTT should measurably drop once contention is reduced (we do not have a specific target threshold without knowing the current channel load, but a drop toward the sub-1 ms range typical of an uncontended wired first hop would confirm it).

F-3: Transient Congestion Spikes at Busy Hour

Saw: During the busy-hour capture, RTT to T3 sat on a ∼27 ms baseline but spiked repeatedly to as high as ∼192 ms, compared to a flat ∼48 ms baseline with only isolated spikes (max ∼155 ms) at quiet hour (Chapter 3, Figure 3.3).

Evidence: E-9, Figure 3.3

Think: We conclude the sharp-in/sharp-out spike shape reflects bursty cross-traffic filling a router buffer and draining quickly once the burst ends, rather than sustained congestion; this is consistent with what Chapter 3 already argues from the same plot. One other explanation we couldn't rule out is that these spikes are ISP-side micro-outages or momentary throttling at the Tata Communications peering edge identified in Chapter 1, rather than a purely local buffer filling: we have no way to distinguish "our packet queued behind others" from "the upstream link itself briefly stalled" without visibility into that router, which is outside what we can observe from a single end host.

Fix: If the cause is local buffering, enabling an active queue management scheme (e.g. fq_codel) on the campus egress gateway would bound queuing delay by dropping/marking early instead of letting the buffer fill; this would not help if the cause is upstream, which is why distinguishing the two matters before recommending a fix with confidence.

Check: Repeat the 300-packet busy-hour capture; under the local-buffering explanation, adding AQM should visibly cap spike height in a later capture. Under the upstream explanation, spikes would persist regardless of any change made on campus-controlled equipment, which would itself be informative.

At least one finding above (F-1) comes from the topology work and at least two (F-2, F-3) come from the delay experiment, per the assignment's requirement of at least one from each.